

**UPDATING CEB'S METHOD FOR
SCREENING-LEVEL ESTIMATES OF DERMAL EXPOSURE
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Chemical Engineering Branch
Economics, Exposure and Technology Division**

The purpose of this update is to revise the surface area parameter default values for one and two hands based on the mean skin surface area for adult males in the September 2011 update of the EPA Exposure Factors Handbook (EFH).

The current Chemical Engineering Branch (CEB) Dermal Exposure Assessment method was established in 2000. The 1997 EPA EFH contained a value of 840 cm² mean skin surface area for adult males' hands, and this value was adopted as the surface area parameter default values for two hands in the 2000 method. The 2011 EPA EFH contains a value of 1,070 cm² mean skin surface area for adult males' hand.

Updating the dermal method to adopt the 2011 EFH mean skin surface area for adult males will result in a surface area parameter default value for two hands of 1,070 cm² and a surface area parameter default value for one hand of 535 cm². The updates are shown in Table 1 below.

Table 1: Factors for Screening-Level Assessments of Dermal Exposure to the Hands

Type of Contact ¹	Typical Examples	S ₂ (cm ²)	Q ³ (mg/cm ²)	Resulting Dermal Contact (mg)
Routine, direct handling of solids - 2 hands	· Filling/dumping containers of powders, flakes, granules · Weighing powder/ scooping/mixing (i.e., dye weighing) · Handling wet or dried material in a filtration and drying process			up to 3,100 ⁴
Routine contact with surfaces - 2 hands - solids	· Handling bags of solid materials (closed or empty)			up to 1,100 ⁴
Routine immersion, 2 hands - liquids	· Handling wet surfaces · Spray painting	1,070	1.3 - 10.3	up to 11,000
Routine contact, 2 hands - liquids	· Maintenance · Manual cleaning of equipment · Filling drum with liquid	1,070	0.7 - 2.1	up to 2,200
Incidental contact, 2 hands - liquids	· Connecting transfer line	1,070	0.7 - 2.1	up to 2,200
Incidental contact, 1 hand - liquids	· Sampling · Ladling liquid/bench scale liquid transfer	535	0.7 - 2.1	up to 1,100

Notes:

1- The terms “routine” and “incidental” reflect typical CEB judgments on likelihood of contact for the example activities.

2 - Values of the skin surface area of the hands taken from: EPA Exposure Factors Handbook, 2011 and are the mean values for men

3 - Selected ranges of ‘Q’ Values for liquid handling activities taken from: EPA, 1992. A Laboratory Method to Determine the Retention of Liquids on the Surface of Hands, Exposure Evaluation Division, Office of Pollution Prevention and Toxic, USEPA, EPA 747-R-92-003, September, 1992.

4 - Values for dermal contact for solids handling activities were taken from: Lansink, 1996. Lansink, C.J.M., M.S.C. Breelen, J. Marquart, and J.J. van Hemmen: Skin Exposure to Calcium Carbonate in the Paint Industry. Preliminary Modeling of Skin Exposure Levels to Powders Based on Field Data (TNO Report V 96.064). Rijswijk, The Netherlands: TNO Nutrition and Food Research Institute, 1996.

Aside from the skin surface area update, further details on derivation of this table can be found in: CEB, 2000. Options for Revising CEB’s Method for Screening-Level Estimates of Dermal Exposure. Final Report. 06/01/00.